

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

< reasoning process here >

To find the de Broglie wavelength of an electron with a kinetic energy of 1 eV , we use the de Broglie wavelength formula :

$\lambda = h / p$

where :

- h is Planck's constant = 6.626×10^{-34} J·s (or 6.626×10^{-27} erg·s)
- p is the momentum of the electron

For non-relativistic electrons , kinetic energy (K.E.) is related to momentum by :

$K.E. = (1/2)mv^2 = p^2 / (2m)$

So , solving for momentum p :

$p = \sqrt{2mK.E.}$

Thus ,

$\lambda = h / \sqrt{2mK.E.}$

We can plug in values to compute the wavelength .

Step 1 : Convert KE to joules .

$1\text{ eV} = 1.602 \times 10^{-19}\text{ J}$

So , $K.E. = 1.602 \times 10^{-19}\text{ J}$

Step 2 : Mass of electron $m = 9.109 \times 10^{-31}\text{ kg}$

Step 3 : Compute momentum :

$p = \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}}$
 $= \sqrt{2 \times 9.109 \times 1.602 \times 10^{-50}}$
 $= \sqrt{2.918 \times 10^{-50}}$
 $= 1.708 \times 10^{-25}\text{ kg}\cdot\text{m/s}$

Step 4 : Compute $\lambda = h / p$

$h = 6.626 \times 10^{-34}\text{ J}\cdot\text{s}$

$\lambda = 6.626 \times 10^{-34} / 1.708 \times 10^{-25} \approx 3.88 \times 10^{-9}\text{ m}$

Convert to cm : $3.88 \times 10^{-9}\text{ m} = 3.88 \times 10^{-7}\text{ cm}$

Now , check the options :

- A. 2.48×10^{-8} cm $\rightarrow$ too small
- B. 5.92×10^7 cm/sec $\rightarrow$ units are speed , not wavelength $\rightarrow$ invalid
- C. 8.66×10^{-6} cm $\rightarrow$ too large
- D. 3.14×10^{-5} cm $\rightarrow$ too large
- E. 9.81×10^{-4} cm $\rightarrow$ too large
- F. 5.27×10^{-9} m $\rightarrow 5.27 \times 10^{-7}$ cm $\rightarrow$ too small
- G. 1.23×10^{-7} cm $\rightarrow 1.23 \times 10^{-9}$ m $\rightarrow$ too small
- H. 4.14×10^{-10} m $\rightarrow 4.14 \times 10^{-8}$ cm $\rightarrow$ too small
- I. 6.625×10^{-27} erg·s $\rightarrow$ units are h , not wavelength
- J. 0.512×10^6 eV $\rightarrow$ energy , not wavelength

But our computed value is approximately $3.88 \times 10^{-9}\text{ m} = 3.88 \times 10^{-7}\text{ cm}$

Wait $\rightarrow$ let's check if we made a unit error .

We found : $\lambda \approx 3.88 \times 10^{-9}\text{ m} = 3.88 \times 10^{-7}\text{ cm}$

But among the options :

F is $5.27 \times 10^{-9}\text{ m} \rightarrow 5.27 \times 10^{-7}\text{ cm}$